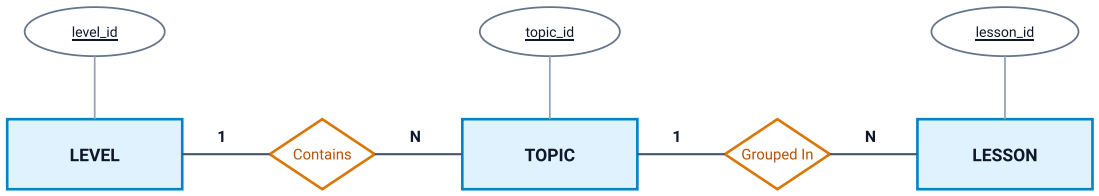
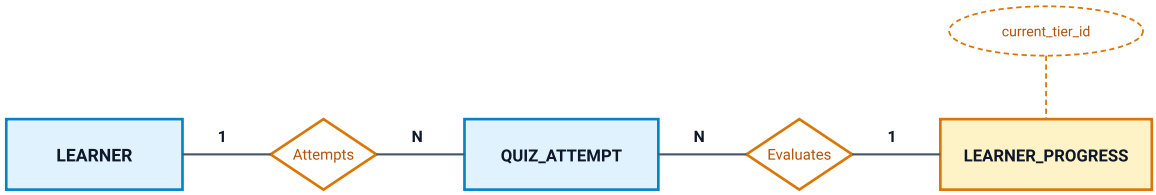
ENTITY RELATIONSHIP DIAGRAMS

# Chen ERDs: Catalogue & Adaptive Assessment

## 1. Learning Catalogue & Assessment Structure



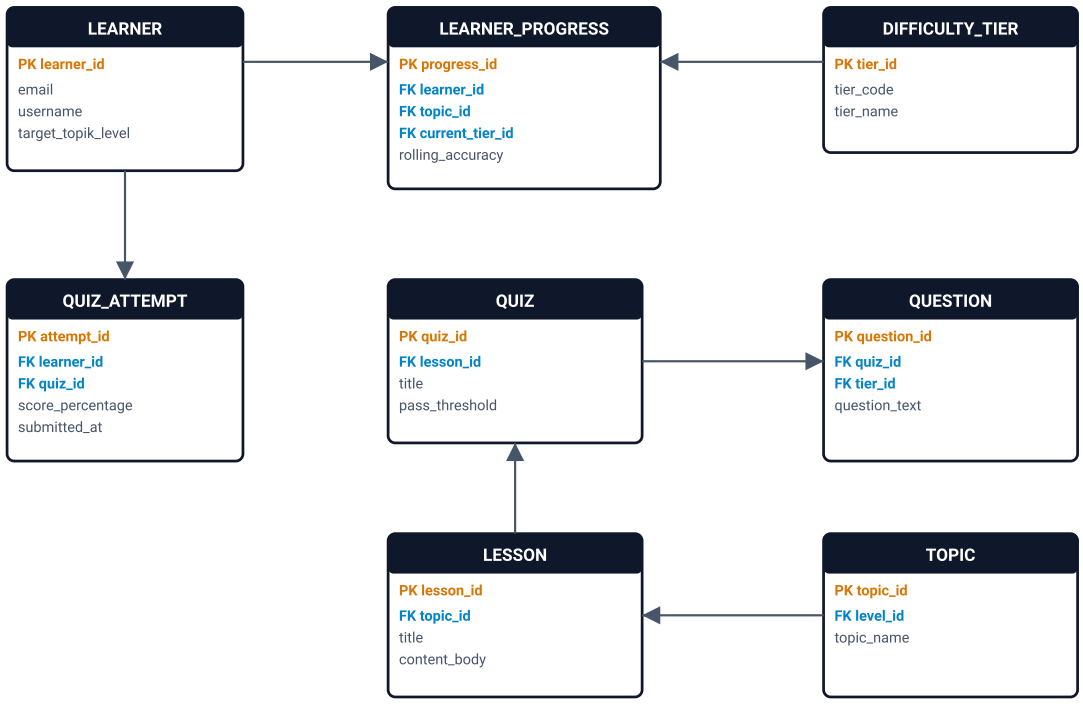

## 2. Adaptive Progress & Assessment History




**Explanation:** Chen notation displays the high-level conceptual hierarchy of learning catalog elements and the adaptive feedback loop connecting student quiz execution to persistent topic progress metrics.

ENTITY RELATIONSHIP DIAGRAMS

# IE (Crow's Foot) Logical Relational Model




**Explanation:** The IE Crow's Foot diagram translates conceptual entities into relational tables with explicit Primary Keys (PK) and Foreign Keys (FK). Crow's feet mark the "many" side of relations, preserving referential integrity across learning catalogs, quiz execution, and progress tracking.